

Applied Bayesian Statistics in Animal & Biological Sciences

3) Linear models & multiple comparisons

Henk J. van Lingen

hvanling@uoguelph.ca

May 12, 2026

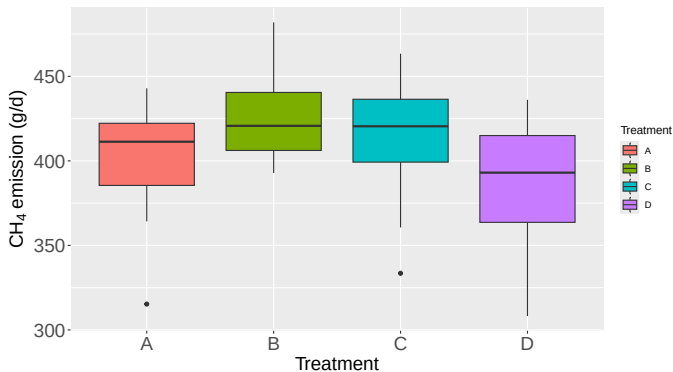


Today's program

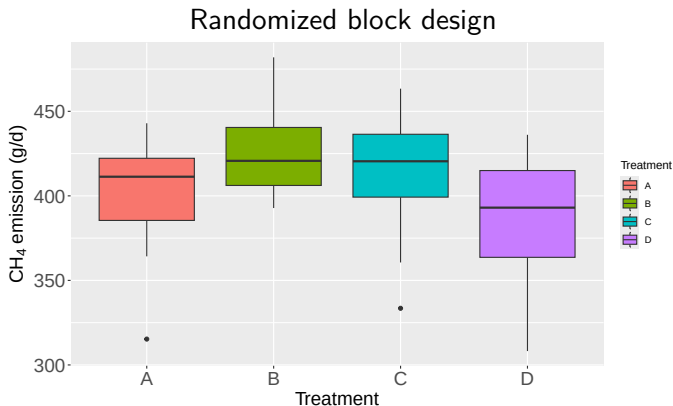
- Rationale behind multiple comparison adjustment
- Multiple comparisons in a Bayesian context
- Model selection metrics in a Bayesian context
- Multiple comparisons through priors (shrinkage effects), model selection and model posterior probability
- (R)STAN code for Bayesian multiple comparisons
- t -distribution for linear model fitting

Multiple comparisons - CH₄ from cattle for 4 diets

Randomized block design



Multiple comparisons - CH₄ from cattle for 4 diets



$$y_{ijk} = \beta_i + u_j + \epsilon_{ijk} \text{ with } u_j \sim N(0, \sigma_u^2) \text{ and } \epsilon_{ijk} \sim N(0, \sigma_e^2) \quad (1)$$

	Diet A	Diet B	Diet C	Diet D	SEM	<i>P</i> -value
CH ₄ (g/d)	399	428	411	386	14.1	0.08

Multiple comparisons in a frequentist context

Contrast	Estimate	SE	P -value	adj P -value
A - B	-29.6	16.0	0.08	0.28
A - C	-12.4	16.0	0.45	0.86
A - D	13.1	16.0	0.42	0.84
B - C	17.2	16.0	0.29	0.71
B - D	42.7	16.0	0.01	0.06
C - D	25.5	16.0	0.12	0.40

- Probability of falsely rejecting at least one H_0 (type I error) increases with the number of comparisons
- Apply *empirical* adjustments (e.g., Tukey, Bonferroni, etc) in favor of H_0

Multiple comparisons in a Bayesian context

Prior effect

- A less informative prior leads to a posterior in favor of H_0
- An increased number of comparisons requires a less informative prior
- Shrinkage: group mean estimates have shrunk towards the overall mean (due to less informative prior)

Multiple comparisons in a Bayesian context

Prior effect

- A less informative prior leads to a posterior in favor of H_0
- An increased number of comparisons requires a less informative prior
- Shrinkage: group mean estimates have shrunk towards the overall mean (due to less informative prior)

Bayesian hierarchical model effect

- Multiplicity adjustment when observed *data* provides evidence for H_0

Multiple comparisons in a Bayesian context

Prior effect

- A less informative prior leads to a posterior in favor of H_0
- An increased number of comparisons requires a less informative prior
- Shrinkage: group mean estimates have shrunk towards the overall mean (due to less informative prior)

Bayesian hierarchical model effect

- Multiplicity adjustment when observed *data* provides evidence for H_0

2 Bayesian approaches:

- Parameter estimation - credibility of parameter posteriors
- Model selection - for strongest evidence favoring H_0

Model selection in Frequentist and Bayesian statistics

- Information criteria quantify the goodness-of-fit $\hat{L}$ penalized for the number of model parameters a : e.g. $AIC = -2\log(\hat{L}) + 2a$
- Preferred model has the lowest score

Model selection in Frequentist and Bayesian statistics

- Information criteria quantify the goodness-of-fit $\hat{L}$ penalized for the number of model parameters a : e.g. $AIC = -2\log(\hat{L}) + 2a$
- Preferred model has the lowest score

Information Criterion	Inference	Full Prob. Dist.
Akaike (AIC)	Frequentist	NA
Bayesian (BIC)	Frequentist	NA
Deviance (DIC)	Bayesian	No
Widely applicable (WAIC)	Bayesian	Yes
Leave-one-out (LOO-IC)	Bayesian	Yes

Model selection in Frequentist and Bayesian statistics

- Information criteria quantify the goodness-of-fit $\hat{L}$ penalized for the number of model parameters a : e.g. $AIC = -2\log(\hat{L}) + 2a$
- Preferred model has the lowest score

Information Criterion	Inference	Full Prob. Dist.
Akaike (AIC)	Frequentist	NA
Bayesian (BIC)	Frequentist	NA
Deviance (DIC)	Bayesian	No
Widely applicable (WAIC)	Bayesian	Yes
Leave-one-out (LOO-IC)	Bayesian	Yes

- WAIC is also called Watanabe-Akaike information criterion

Bayesian model selection through WAIC and LOOIC

Log pointwise predictive density - for WAIC - n data points

$$\sum_{i=1}^n \log p(y_i|y) = \sum_{i=1}^n \log \int p(y_i|\theta)p(\theta|y)d\theta \quad (2)$$

Vehtahri et al., 2017

Bayesian model selection through WAIC and LOOIC

Log pointwise predictive density - for WAIC - n data points

$$\sum_{i=1}^n \log p(y_i|y) = \sum_{i=1}^n \log \int p(y_i|\theta)p(\theta|y)d\theta \quad (2)$$

Leave-one-out log-pointwise predictive density

$$\sum_{i=1}^n \log p(y_i|y_{-i}) = \sum_{i=1}^n \log \int p(y_i|\theta)p(\theta|y_{-i})d\theta \quad (3)$$

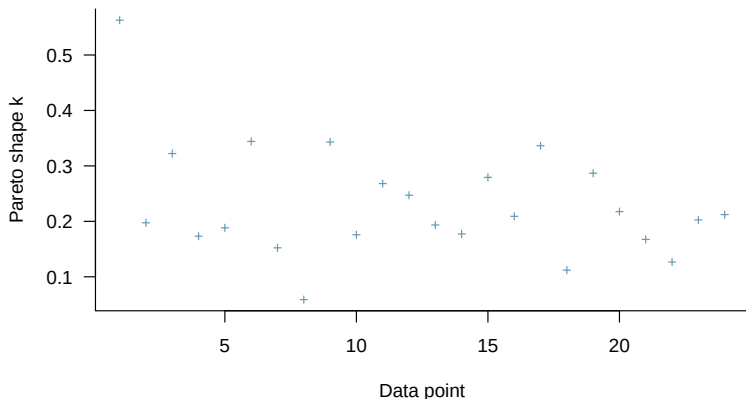
- Fit a pareto distribution to the 20% largest importance ratios:

$$r_i^s \propto \frac{p(\theta|y_{-i})}{p(\theta|y)}$$

Vehtahri et al., 2017

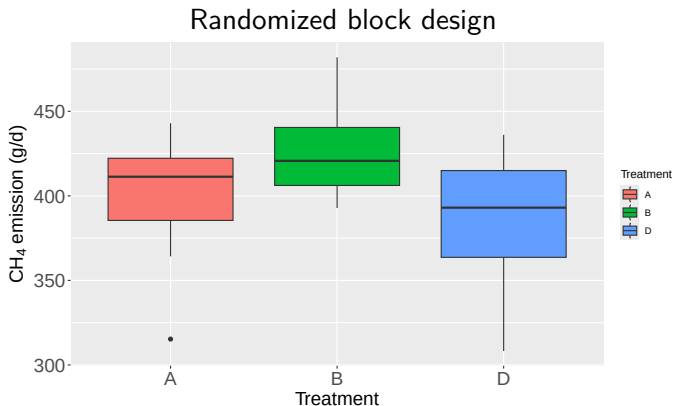
Evaluate the $\hat{k}$ shape parameters of the pareto distributions

Pareto diagnostic plot



- Perfect: $\hat{k} < 0.5$; Acceptable: $\hat{k} < 0.7$; STAN warns you: $\hat{k} \geq 0.7$;
Unacceptable: $\hat{k} \geq 1$

Multiple comparisons - CH₄ from cattle for 3 diets



$$y_{ijk} = \beta_i + u_j + \epsilon_{ijk} \text{ with } u_j \sim N(0, \sigma_u^2) \text{ and } \epsilon_{ijk} \sim N(0, \sigma_e^2) \quad (4)$$

	Diet A	Diet B	Diet D	SEM	P-value
CH ₄ (g/d)	399	428	386	13.7	0.03

Multiple comparisons and the Bell number - 3 diets

Diet	M_1	M_2	M_3	M_4	M_5
A	1	1	1	1	1
B	1	2	1	2	2
D	1	1	2	2	3

- Hypothesize diets have the same or different CH₄ emissions
- Number of partitions for a set of I elements is known as the Bell number (B_I), $B_3 = 5$
- Fit the 5 models, then compute likelihood and AIC, BIC (Frequentist models), or DIC, WAIC, LOOIC (Bayesian models)

Neath et al., 2018

Bayesian posterior probability of evaluated models

Non-informative priors

$$\beta \sim N(400, 10^4), \sigma_b \sim N_+(0, 10^4) \text{ and } \sigma_e \sim N_+(0, 10^4)$$

Bayesian posterior probability of evaluated models

Non-informative priors

$$\beta \sim N(400, 10^4), \sigma_b \sim N_+(0, 10^4) \text{ and } \sigma_e \sim N_+(0, 10^4)$$

Approximation of posterior probability of model k out of L models

$$\pi(M_k|y_N) = \frac{\exp(-\Delta B_k/2)}{\sum_{l=1}^L \exp(-\Delta B_l/2)} \quad (5)$$

Model k	H_0	LOOIC	ΔB_k	$\pi(M_k y_N)$
M_1	$\mu_A = \mu_B = \mu_D$	248.6	4.86	0.05
M_2	$\mu_A = \mu_D, \mu_B$	243.7	0.00	0.51
M_3	$\mu_A = \mu_B, \mu_D$	246.6	2.88	0.12
M_4	$\mu_A, \mu_B = \mu_D$	251.6	7.93	0.01
M_5	μ_A, μ_B, μ_D	244.7	1.00	0.31

Bayesian posterior probability of evaluated models

Informative priors

$$\beta \sim N(400, 40), \sigma_b \sim N_+(0, 16) \text{ and } \sigma_e \sim N_+(0, 8)$$

Bayesian posterior probability of evaluated models

Informative priors

$$\beta \sim N(400, 40), \sigma_b \sim N_+(0, 16) \text{ and } \sigma_e \sim N_+(0, 8)$$

Model k	H_0	LOOIC	ΔB_k	$\pi(M_k y_N)$
M_1	$\mu_A = \mu_B = \mu_D$	249.7	4.94	0.04
M_2	$\mu_A = \mu_D, \mu_B$	244.7	0.00	0.44
M_3	$\mu_A = \mu_B, \mu_D$	246.2	1.51	0.21
M_4	$\mu_A, \mu_B = \mu_D$	251.2	6.45	0.02
M_5	μ_A, μ_B, μ_D	245.6	0.84	0.29

- No major changes, data-driven final outcome

Bayesian posterior probability for pairwise mean equalities

Hypothesis testing for 3 treatments

Hypothesis	P (non-inf prior)	P (inf prior)
$\mu_A = \mu_B$	$0.05 + 0.12 = 0.17$	$0.04 + 0.21 = 0.25$
$\mu_A = \mu_D$	$0.05 + 0.51 = 0.56$	$0.04 + 0.44 = 0.48$
$\mu_B = \mu_D$	$0.05 + 0.01 = 0.06$	$0.04 + 0.02 = 0.06$

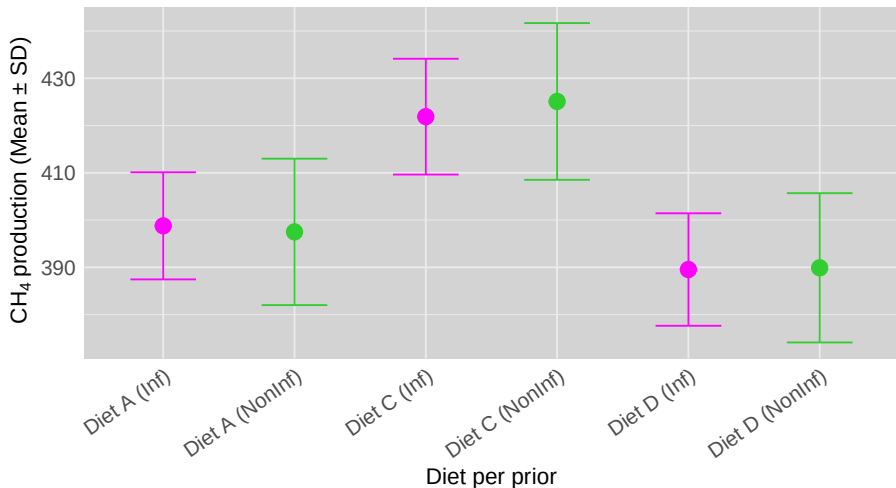
Estimated groups means under Bayesian Model Averaging

Non-informative prior

Model k	β_A	β_B	β_D	$\pi(M_k y_N)$
M_1	404±13.8	404±13.8	404±13.8	0.05
M_2	392±14.9	428±17.0	392±14.9	0.51
M_3	414±14.9	414±14.9	386±17.4	0.12
M_4	407±14.9	399±17.9	399±17.9	0.01
M_5	399±17.0	428±16.8	385±16.9	0.31
BMA	397±15.5	425±16.6	390±15.8	

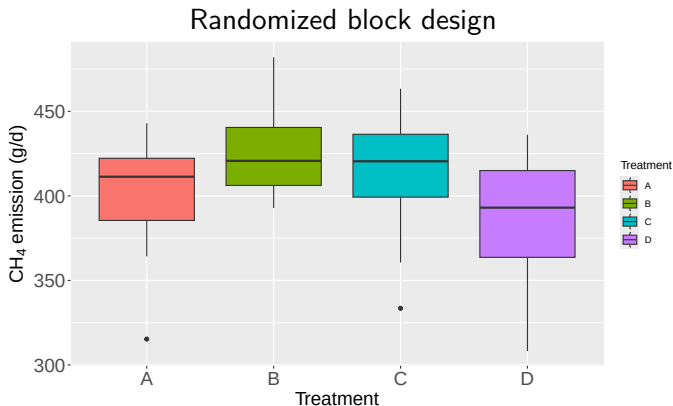
- Example for BMA of diet A: $0.05 \times 404 + 0.51 \times 392 + 0.12 \times 414 + 0.01 \times 407 + 0.31 \times 399 = 397$
- Report BMA values when publishing results

Bayesian Model Averaging for (non)-informative priors



- Shrinkage towards zero-effect on larger standard deviations for non-informative prior

Multiple comparisons - CH₄ from cattle for 4 diets



$$y_{ijk} = \beta_i + u_j + \epsilon_{ijk} \text{ with } u_j \sim N(0, \sigma_u^2) \text{ and } \epsilon_{ijk} \sim N(0, \sigma_e^2) \quad (6)$$

	Diet A	Diet B	Diet C	Diet D	SEM	<i>P</i> -value
CH ₄ (g/d)	399	428	411	386	14.1	0.08

Multiple comparisons and the Bell number - 4 diets

Diet	M_1	M_2	M_3	M_4	M_5	M_6	M_7	...	M_{13}	M_{14}	M_{15}
A	1	2	1	1	1	1	1	...	1	1	1
B	1	1	2	1	1	1	2	...	2	2	2
C	1	1	1	2	1	2	2	...	2	3	3
D	1	1	1	1	2	2	1	...	3	3	4

- Hypothesize diets have the same or different CH_4 emissions
- Number of partitions for $I = 4$ elements now yields $B_4 = 15$
- Note: $B_5 = 52$ and $B_6 = 203$...

Bayesian posterior probability of evaluated models

Approximation of posterior probability of model k out of L models

$$\pi(M_k|y_N) = \frac{\exp(-\Delta B_k/2)}{\sum_{l=1}^L \exp(-\Delta B_l/2)} \quad (7)$$

Model k	H_0	LOOIC	B_k	$\pi(M_k y_N)$
M_1	$\mu_A = \mu_B = \mu_C = \mu_D$	329.5	3.55	0.03
M_2	$\mu_A = \mu_C = \mu_D, \mu_B$	332.0	6.03	0.01
M_3	$\mu_A = \mu_B = \mu_D, \mu_C$	327.7	1.74	0.08
M_4	$\mu_A = \mu_B = \mu_C, \mu_D$	326.0	0.08	0.18
M_5	$\mu_A = \mu_B = \mu_C, \mu_D$	331.8	5.92	0.01
M_6	$\mu_A = \mu_B, \mu_C = \mu_D$	325.9	0.00	0.19
...				
M_{15}	$\mu_A, \mu_B, \mu_C, \mu_D$	329.0	3.11	0.04

Bayesian posterior probability for pairwise mean equalities

Hypothesis testing for 4 treatments

Hypothesis	P (Bayes)	P (Tukey)
$\mu_A = \mu_B$	0.16	0.28
$\mu_A = \mu_C$	0.46	0.86
$\mu_A = \mu_D$	0.53	0.84
$\mu_B = \mu_C$	0.38	0.71
$\mu_B = \mu_D$	0.06	0.06
$\mu_C = \mu_D$	0.30	0.40

- Same conclusion for Bayes and Tukey in this case: no differences if testing at $\alpha = 0.05$, μ_B and μ_D tend to be different

Hierarchical model for randomized block design in STAN

```
model_rbd <- "  
data {  
  int<lower=1> N; // number of data points  
  int<lower=1> no_trt;  
  int trt[N]; // the treatments  
  int<lower=1> no_block; //  
  int block[N]; // the variance groups  
  vector[N] y; // the data  
  real sd_beta; real sd_b; real sd_e; // priors: sd of beta, b groups and e  
} // end of data  
  
parameters{  
  vector[no_trt] beta; // treatment means  
  vector[no_block] blk; //  
  real<lower=0> sigma_b; // block std dev  
  real<lower=0> sigma_e; // res std dev  
} // end of parameters  
  
model {  
  beta ~ normal(400,sd_beta); blk ~ normal(0,sigma_b);  
  sigma_b ~ normal(0, sd_b); // These are half-normal priors  
  sigma_e ~ normal(0, sd_e); // These are half-normal priors  
  
  y ~ normal(beta[trt] + blk[block], sigma_e);  
} // End of model  
.  
.  
.
```

Hierarchical model and LOOIC in STAN

- Add generated quantities block to compute the log-likelihood:

```
generated quantities {
  vector[N] log_lik;
  vector[N] mu = beta[trt] + blk[block];

  // to use loo package and compute waic - MUST be called log_lik
  for (n in 1:N){
    log_lik[n] = normal_lpdf(y[n] | mu[n], sigma_e);
  } // End of for loop n
} // End of generated quantities
"
```

Hierarchical model and LOOIC in STAN

- Add generated quantities block to compute the log-likelihood:

```
. . .  
  
generated quantities {  
  vector[N] log_lik;  
  vector[N] mu = beta[trt] + blk[block];  
  
  // to use loo package and compute waic - MUST be called log_lik  
  for (n in 1:N){  
    log_lik[n] = normal_lpdf(y[n] | mu[n], sigma_e);  
  } // End of for loop n  
  
} // End of generated quantities  
"
```

- Code `beta[trt] + blk[block]` outside `normal_lpdf( ... )`

Hierarchical model for randomized block design in RSTAN

```
### Load rstan package
library(rstan)
options(mc.cores = parallel::detectCores())
rstan_options(auto_write = TRUE)

### Load loo package for computing LOOIC values from rstan model outputs
library(loo)

### Initialize STAN code in R
comp_mod_rbd <- stan_model(model_code=model_rbd,model_name="model_rbd")

### Load group assignment code and load data
source("Group_assgn.R")
CH4_dat <- read.csv("CH4_K402.csv")

### Load partitions package
library(partitions)

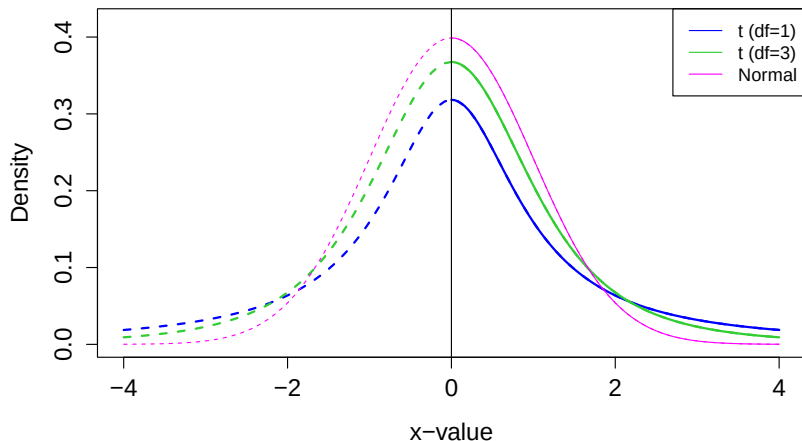
### Design Bell matrix/hypotheses
max.grps <- length(unique(CH4_dat$Treatment))
parts <- setparts(max.grps)
trt_mat <- apply(parts, 2, grp.asn, npg=table(CH4_dat$Treatment))

. . .
```


Run hierarchical model and extract LOOIC in RSTAN

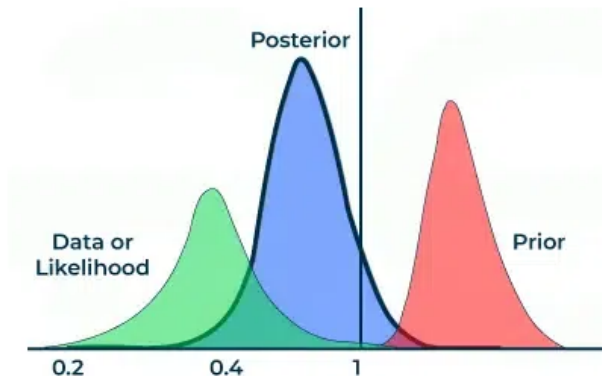
```
...  
  
### Prepare STAN data list object  
reg.dat <- list(N=nrow(CH4_dat), trt=trt_mat[,15], no_trt=max(trt_mat[,15]),  
               block=CH4_dat$Period, no_block=length(unique(CH4_dat$Period)),  
               y=CH4_dat$CH4)  
  
CH4_fit.rs <- sampling(comp_mod_rbd, data=reg.dat, iter=5e3,  
                      pars=c("beta", "blk", "sigma_b", "sigma_e", "log_lik"),  
                      warmup=1e3, thin=25, chains=4)  
  
print(CH4_fit.rs, pars=c("beta", "blk", "sigma_b", "sigma_e"))  
  
# Extract log-likelihood  
LL <- extract_log_lik(CH4_fit.rs, parameter_name="log_lik", merge_chains=F)  
r_eff <- relative_eff(LL) # r_eff <- relative_eff(exp(LL))  
lootest <- loo(LL, r_eff=r_eff, moment_match = TRUE)  
  
print(lootest$estimates)  
  
##           Estimate           SE  
## elpd_loo -163.70748  5.383638  
## looic    327.41497 10.767275
```

Reconsider a t -distributed prior for heavier tails?



- Stan lets you analyse designed experiments as usual, but gives full probability statements for any effect
- Stan avoids software-imposed simplifications and allows more flexible models, e.g. $\sigma_b \sim t_+(\nu, 0, 16)$

Thank you for your attention!



Questions?

Exercise lecture 3

- Reproduce the rbd STAN code presented (Dataset on GitHub)
- Write a STAN model for a (3×3) Latin square design (Dataset on GitHub)
- Change or extend your STAN models with different priors:
 - Change prior parameters
 - Take other prior distributions
- Simulate a dataset (e.g RBD, LSq, etc) if needed and fit your Bayesian model
- Compare your results with frequentist multiple comparisons